

Study on vortex-induced vibration mitigation of parallel bridges by multi-tuned mass damper inerter

Wei Lei^b, Qi Wang^{a,b,*}

^a Wind Engineering Key Laboratory of Sichuan Province, Southwest Jiaotong University, Chengdu 610031, China

^b School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, China

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ABSTRACT

This study proposes a novel Multi-Tuned Mass Damper Inerter (MTMDI) system for mitigating the vortex-induced vibration (VIV) responses of parallel bridges. In the proposed system, two individual bridges are linked using two Tuned Mass Damper Inerters (TMDIs). A model of parallel bridges with MTMDI under vortex-induced force is constructed, and two additional schemes wherein each bridge is equipped with a TMDI and a Tuned Mass Damper (TMD) separately are considered. The VIV responses of the parallel bridges are estimated through full bridge aeroelastic model wind tunnel tests. The performance of the MTMDI system with the optimal parameters obtained by the multi-objective optimization technique is compared with the performance of the other aforementioned two schemes. The results indicate that the optimal MTMDI system with smaller attached mass and inertance effectively mitigates the vertical VIV responses when applied to the parallel bridges. The frequency response functions and displacement time histories further demonstrate the effectiveness of the MTMDI system as an attractive alternative to the other schemes. Finally, the sensitivity of performance against variations in each parameter in the MTMDI system is discussed.

1. Introduction

Vortex-induced vibrations (VIVs) are a kind of vibrations typically observed in long-span bridges under wind loads, and are more likely to occur due to the decrease in bridge damping over long-term operation and changes in aerodynamic shape [1,2]. For example, some researchers believe that the vibrations of the Humen Bridge [3] is caused by VIVs resulting from temporary water-filled barriers which alter the aerodynamic shape of the girder. In addition, long-span bridges are characterized by low damping, flexibility, and natural frequency, and the auxiliary elements of the bridges, such as railings and traffic barriers, contribute to girder bluntness. Periodical vortex shedding occurs when air flows pass these elements and girders. VIVs are usually observed when the vortex shedding frequency is close to the natural frequency of the bridge. VIVs typically exhibit limited vibration amplitudes, in contrast with divergent flutter and galloping. However, long-term large-amplitude VIVs can lead to avoidable issues for the bridges, including fatigue failure, structural damage, and compromised driving comfort [4,5].

Currently, the aerodynamic performance of girders is generally improved by modifying the location or size of the affiliated facilities,

such as handrails [6], wind barriers [7], maintenance trails [8], and traffic barriers [9]. Additional components, including guide vanes and spoilers [10], grid plates [11], wind fairing, and guide plates [12], and upper or lower central stabilizer plates [13] have been successfully applied as measures for enhanced aerodynamic performance. The layouts of extant aerodynamic measures on the girders were summarized by Fujino and Siringoringo [14], providing valuable references for determining appropriate countermeasures in wind tunnel tests. Computational fluid dynamics (CFD) [15] and field visualization techniques [16] are often used to observe vortex separation, shedding, and reattachment near the main girder, affording insights into the VIV mechanism. The aforementioned aerodynamic measures are conducive to VIV suppression of the main girder. However, they usually require multiple rounds of wind tunnel tests or numerical simulations. Some aerodynamic measures are complex and involve difficulties in installation and maintenance. Moreover, there is no universally applicable form of aerodynamic measures for all types of main girders. Additionally, these measures are required to simultaneously suppress VIVs and flutter [17]. This poses further challenges in designing the aerodynamic configurations of long-span bridges. Another approach to mitigating VIVs is the installation of mechanical control devices to alter dynamic

* Corresponding author at: Wind Engineering Key Laboratory of Sichuan Province, Southwest Jiaotong University, Chengdu 610031, China.

E-mail address: wangchee_wind@swjtu.edu.cn (Q. Wang).

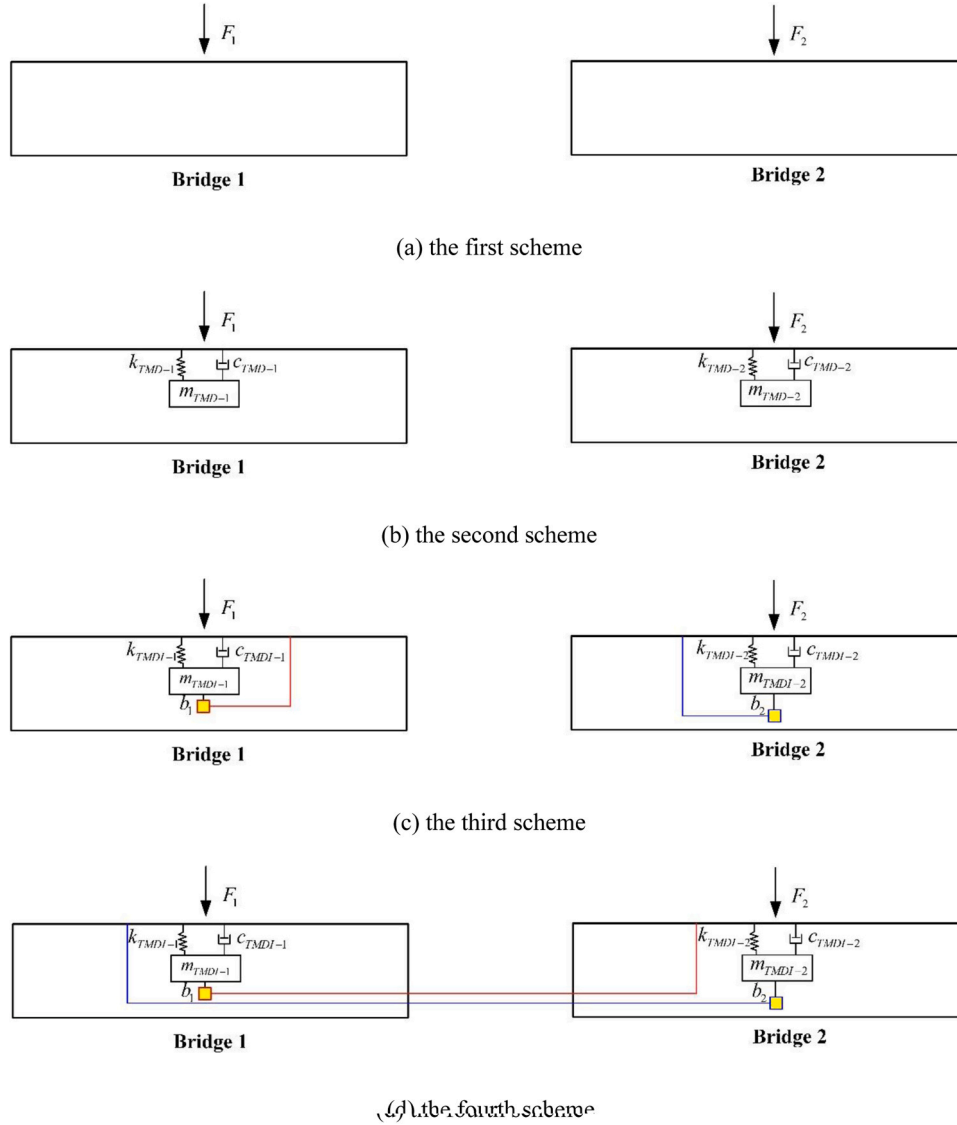


Fig. 1. Four schemes of the parallel bridges: (a) original structure; (b) two TMDs in each bridge (separately); (c) two TMDIs in each bridge (separately); (d) two TMDIs-linked bridge1 and bridge 2.

characteristics, such as Tuned Mass Damper (TMD) [18–21]. By adjusting the mass ratio, frequency ratio, and damping ratio of TMD, a large amount of vibration energy from the structure can be absorbed and dissipated. TMD and its other variants, such as Pounding Tuned Mass Damper (PTMD) [22] and Multiple Tuned Mass Damper (MTMD) [23], are also widely used in buildings, chimneys, and other engineering structures and can substantially mitigate wind-induced and seismic responses [24,25]. However, TMD requires a relatively large mass to achieve notable vibration control effects. The large static stretching of springs at low frequencies significantly hinders their wide application in bridges [26], and the height limitations of streamlined steel box girders often restrict the installation of large-volume TMD. For example, the TMD devices installed on the Hong Kong-Zhuhai-Macao Bridge has to be light-weight [27].

In recent years, a novel control device, namely, Tuned Mass Damper Inerter (TMDI), is developed [28–30] and has been widely used for wind [31,32] and earthquake resistant design [33,34]. Zhu et al. [35] analyzed the effectiveness of the application of TMDI with regard to the wind resistance of linked buildings. The results indicated that TMDI, compared with the traditional TMD, assisted in suppressing the acceleration responses of the two buildings for all wind direction angles. This

was achieved via the amplification of the physical mass due to the use of the inerter—a mass-amplification effect—in the case of the TMDI. Consequently, the TMDI can achieve comparable control performance to those of a TMD with a smaller attached mass [36,37]. The smaller size of the control device not only reduces manufacturing costs but also helps address space limitations when installing TMDI in steel box girders. Better control performance can be achieved when the TMDI topology spans more floors or longer girder lengths [38,39]. These characteristics of TMDI make it prospective in its application in the vibration suppression of structures. Xu et al. [40] introduced TMDI to suppress VIVs. The VIV responses of the main girder were reduced by the optimized TMDI parameters, thus providing an alternative for the VIV suppression of bridges. Xu et al. [41] proposed closed-form design formulas for TMDI, which are suitable for the VIV suppression of bridges. Furthermore, the Multi-Tuned Mass Damper Inerter (MTMDI) was proposed for the wind [42] and earthquake resistant design [43] of linked buildings. Dai et al. [44] constructed a MTMDI system composed of a three-component TMD and an inerter, and they reported better suppression effects by using optimal parameters as compared with TMD and three-element TMD systems in terms of transient amplitude mitigation, steady-state amplitude mitigation, stroke limitation, and static

stretching reduction. Xu et al. [45] presented an efficient method for multi-mode VIV suppression of long-span bridges through spatially distributed TMDIs (DTMDIs). For parallel bridges, where VIVs have been observed in the field [46,47], VIV mitigation is realized using mechanical measures such as the installation of TMD or MTMD. Compared to TMD, TMDI has better spatial feasibility, absorption, and dissipation performance while having smaller masses. However, to the best of our knowledge, the application of MTMDI for VIV suppression in parallel bridges has not been explored.

$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{0} & \boldsymbol{\alpha}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 & \mathbf{0} & \boldsymbol{\alpha}_2 \\ \boldsymbol{\alpha}_1^T & \mathbf{0} & \mathbf{M}_{\text{TMDI-1}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\alpha}_2^T & \mathbf{0} & \mathbf{M}_{\text{TMDI-2}} \end{bmatrix} \begin{Bmatrix} \ddot{\mathbf{Y}}_1(t) \\ \ddot{\mathbf{Y}}_2(t) \\ \ddot{\mathbf{Y}}_{\text{TMDI-1}}(t) \\ \ddot{\mathbf{Y}}_{\text{TMDI-2}}(t) \end{Bmatrix} + \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} & \boldsymbol{\beta}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_2 & \mathbf{0} & \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_1^T & \mathbf{0} & \mathbf{C}_{\text{TMDI-1}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\beta}_2^T & \mathbf{0} & \mathbf{C}_{\text{TMDI-2}} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{Y}}_1(t) \\ \dot{\mathbf{Y}}_2(t) \\ \dot{\mathbf{Y}}_{\text{TMDI-1}}(t) \\ \dot{\mathbf{Y}}_{\text{TMDI-2}}(t) \end{Bmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{0} & \boldsymbol{\gamma}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 & \mathbf{0} & \boldsymbol{\gamma}_2 \\ \boldsymbol{\gamma}_1^T & \mathbf{0} & \mathbf{K}_{\text{TMDI-1}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\gamma}_2^T & \mathbf{0} & \mathbf{K}_{\text{TMDI-2}} \end{bmatrix} \begin{Bmatrix} \mathbf{Y}_1(t) \\ \mathbf{Y}_2(t) \\ \mathbf{Y}_{\text{TMDI-1}}(t) \\ \mathbf{Y}_{\text{TMDI-2}}(t) \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_1(t) \\ \mathbf{F}_2(t) \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix} \quad (2)$$

In this study, MTMDI is adopted as an innovative solution for the VIV control of parallel bridges. The primary focus is on the vertical VIVs observed in wind tunnel tests conducted on a full bridge aeroelastic model of parallel cable-stayed bridges. The optimal MTMDI parameters are obtained using a multi-objective optimization technique. A comparison of the VIV responses among four different control schemes is performed in terms of frequency response functions and displacement time histories. Finally, the robustness of the MTMDI is analyzed by exerting permutations on parameters.

2. Analytical model

A MTMDI model is adopted to suppress the VIVs of parallel bridges. The dynamic characteristics of the system is assumed to be linear. Because the VIVs of bridges usually occurs in a single mode of vertical or torsional, single-frequency vibrations are investigated through the analytical model in this study. An equal and opposite externally applied force F , which is equivalent to an internally developed force, acts on both ends of the inerter equipment. By definition, Eq. (1) holds for an idealized linear inerter [48,49].

$$F = b(\ddot{u}_1 - \ddot{u}_2) \quad (1)$$

where $\ddot{u}_1$ and $\ddot{u}_2$ are the acceleration values of the two ends, and b is the inertance coefficient value.

Four schemes are constructed in this study, as shown in Fig. 1. Fig. 1(a) illustrates the layout of the parallel bridges, that is, the original structure, wherein the vortex-induced forces applied on Bridges 1 and 2 are F_1 and F_2 , respectively. Fig. 1(b) depicts the layout of two separate TMDs installed on the parallel bridges. $m_{\text{TMD-1,2}}$, $c_{\text{TMD-1,2}}$, and $k_{\text{TMD-1,2}}$ denote the masses, damping coefficients of the viscous dampers, and stiffnesses of springs in TMD-1 and TMD-2, respectively. As shown in Fig. 1(c) and (d), the parallel bridges are equipped with a linear TMDI attached to the bridge via a linear spring of stiffness k_{TMDI} and a viscous damper with damping coefficient c_{TMDI} , respectively. Additionally, the TMDI mass m_{TMDI} is directly connected to a linear inerter with an inertance coefficient b . Particularly, in the third scheme, TMDI-1 is installed on Bridge 1, with one end of the inerter b_1 attached to the mass of TMDI-1 and the other end of the inerter b_1 attached to bridge 1. Similarly, TMDI-2 is installed on Bridge 2 with one end of the inerter b_2 attached to the mass of TMDI-2 and the other end of the inerter b_2 attached to Bridge 2. In the fourth scheme, TMDI-1 is installed on Bridge 1, with one end of the inerter b_1 attached to the mass of TMDI-1 and the other end of the

inerter b_1 attached to Bridge 2. Similarly, TMDI-2 is installed on Bridge 2 with one end of the inerter b_2 attached to the mass of TMDI-2 and the other end of the inerter b_2 attached to Bridge 1. $m_{\text{TMDI-1,2}}$, $c_{\text{TMDI-1,2}}$, and $k_{\text{TMDI-1,2}}$ are the masses, damping coefficients, and stiffnesses of TMDI-1 and TMDI-2, respectively.

The equations of vertical VIVs for parallel bridges equipped with TMDIs in the third scheme can be written as

where $\ddot{\mathbf{Y}}_1(t)$, $\ddot{\mathbf{Y}}_2(t)$, $\ddot{\mathbf{Y}}_{\text{TMDI-1}}(t)$, and $\ddot{\mathbf{Y}}_{\text{TMDI-2}}(t)$ represent the acceleration responses of Bridge 1, Bridge 2, TMDI-1, and TMDI-2, respectively. $\dot{\mathbf{Y}}_1(t)$, $\dot{\mathbf{Y}}_2(t)$, $\dot{\mathbf{Y}}_{\text{TMDI-1}}(t)$, and $\dot{\mathbf{Y}}_{\text{TMDI-2}}(t)$ represent the corresponding velocity responses. $\mathbf{Y}_1(t)$, $\mathbf{Y}_2(t)$, $\mathbf{Y}_{\text{TMDI-1}}(t)$, and $\mathbf{Y}_{\text{TMDI-2}}(t)$ represent the respective displacements responses. $\mathbf{F}_1(t)$ and $\mathbf{F}_2(t)$ signify the aerodynamic vortex-induced force vectors applied to Bridges 1 and 2, respectively. Because this study aims to investigate the control effects of the installed TMDI on the parallel bridges, a relatively simple vortex-induced force model, namely, the harmonic force model, is adopted as follows [50]:

$$F_{1,2} = 0.5\rho U B C_{L1,2} \sin(\omega_{s1,2}t + \phi_{1,2}) \quad (3)$$

where ρ is the fluid density, U is the free-stream velocity of the flow, and B is the width of the bridge girder. $C_{L1,2}$, obtained from the experimental results, denote the lift coefficients of Bridge 1 and Bridge 2, respectively, and $\omega_{s1,2}$ denote the vortex shedding circle frequencies and $\phi_{1,2}$ represent the phase differences between the fluid force and displacement of the bridge corresponding to Bridge 1 and Bridge 2, respectively.

For the third scheme, the mass submatrices of the bridge system are obtained as follows

$$\begin{aligned} \mathbf{M}_1 &= \mathbf{M}_1^s + b_1 \\ \mathbf{M}_2 &= \mathbf{M}_2^s + b_2 \\ \mathbf{M}_{\text{TMDI-1}} &= m_{\text{TMDI-1}} + b_1 \\ \mathbf{M}_{\text{TMDI-2}} &= m_{\text{TMDI-2}} + b_2 \\ \boldsymbol{\alpha}_1 &= -b_1 \\ \boldsymbol{\alpha}_2 &= -b_2 \end{aligned} \quad (4)$$

where $\mathbf{M}_1^s$ and $\mathbf{M}_2^s$ denote the mass matrices of Bridges 1 and 2, respectively. $b_{1,2}$ and $m_{\text{TMDI-1,2}}$ denote the inertance coefficients (apparent masses) and masses of TMDI-1 and TMDI-2, respectively. The stiffness sub-matrices of the bridge system are obtained as follows:

$$\begin{aligned} \mathbf{K}_1 &= \mathbf{K}_1^s + k_{\text{TMDI-1}} \\ \mathbf{K}_2 &= \mathbf{K}_2^s + k_{\text{TMDI-2}} \\ \mathbf{K}_{\text{TMDI-1,2}} &= k_{\text{TMDI-1,2}} \\ \boldsymbol{\gamma}_1 &= -k_{\text{TMDI-1}} \\ \boldsymbol{\gamma}_2 &= -k_{\text{TMDI-2}} \end{aligned} \quad (5)$$

where $\mathbf{K}_1^s$ and $\mathbf{K}_2^s$ denote the stiffness matrices of Bridges 1 and 2, and

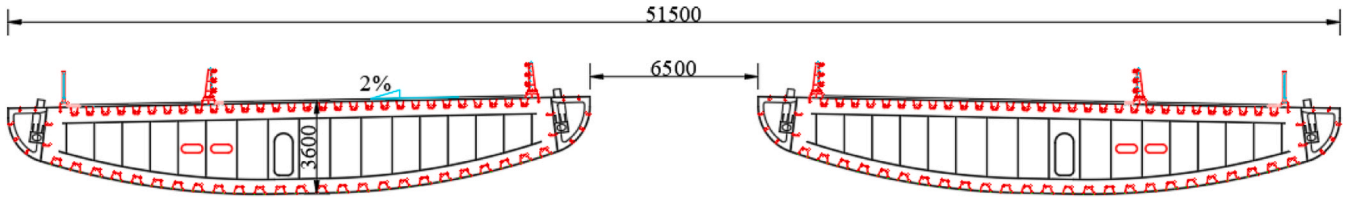
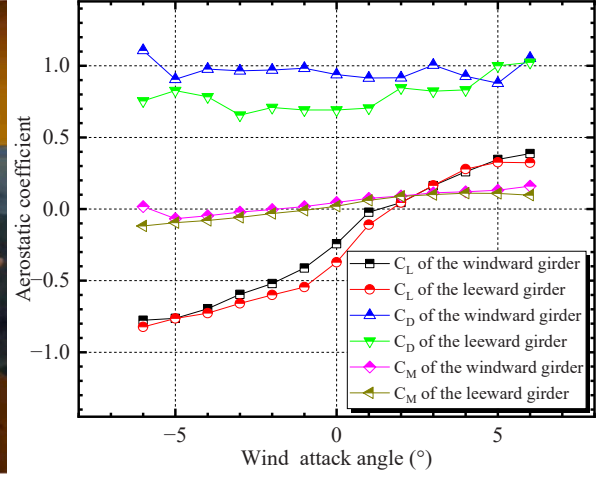


Fig. 2. Configuration of parallel girders section (unit: mm).



(a)



(b)

Fig. 3. Static tests model and results: (a) section model for aerostatic tests at the XNJD-1 wind tunnel (the windward girder); (b) variations in aerostatic coefficients with different attack angles.

$k_{TMDI-1,2}$ denote the stiffness of TMDI-1 and TMDI-2, respectively. The damping sub-matrices of the bridge system are obtained as follows:

$$\begin{aligned} \mathbf{C}_1 &= \mathbf{C}_1^s + c_{TMDI-1} \\ \mathbf{C}_2 &= \mathbf{C}_2^s + c_{TMDI-2} \\ \mathbf{C}_{TMDI-1,2} &= c_{TMDI-1,2} \\ \beta_1 &= -c_{TMDI-1} \\ \beta_2 &= -c_{TMDI-2} \end{aligned} \quad (6)$$

where $\mathbf{C}_1^s$ and $\mathbf{C}_2^s$ represent the damping coefficients of Bridges 1 and 2, and $c_{TMDI-1,2}$ represent the damping coefficients of TMDI-1 and TMDI-2, respectively.

The equations of the vertical VIVs for the bridge system with the installed TMDIs corresponding to the fourth scheme can be expressed as

The difference between Eq. (7) and Eq. (2) is associated with only the mass matrix without variations in the stiffness and damping matrices. The mass submatrices of the bridge system are expressed as follows:

$$\begin{aligned} \mathbf{M}_1 &= \mathbf{M}_1^s + b_2 \\ \mathbf{M}_2 &= \mathbf{M}_2^s + b_1 \\ \mathbf{M}_{TMDI-1} &= m_{TMDI-1} + b_1 \\ \mathbf{M}_{TMDI-2} &= m_{TMDI-2} + b_2 \\ \alpha_1 &= -b_1 \\ \alpha_2 &= -b_2 \end{aligned} \quad (8)$$

Notably, by setting the inertance $b = 0$ in Eq. (8) for the third and the fourth schemes, the equations of motion for the TMDI system degrade to the conventional TMD system, that is, the second scheme. Thus, the

$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{0} & \mathbf{0} & \alpha_2 \\ \mathbf{0} & \mathbf{M}_2 & \alpha_1 & \mathbf{0} \\ \mathbf{0} & \alpha_1^T & \mathbf{M}_{TMDI-1} & \mathbf{0} \\ \alpha_2^T & \mathbf{0} & \mathbf{0} & \mathbf{M}_{TMDI-2} \end{bmatrix} \begin{Bmatrix} \ddot{\mathbf{Y}}_1(t) \\ \ddot{\mathbf{Y}}_2(t) \\ \ddot{\mathbf{Y}}_{TMDI-1}(t) \\ \ddot{\mathbf{Y}}_{TMDI-2}(t) \end{Bmatrix} + \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} & \beta_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_2 & \mathbf{0} & \beta_2 \\ \beta_1^T & \mathbf{0} & \mathbf{C}_{TMDI-1} & \mathbf{0} \\ \mathbf{0} & \beta_2^T & \mathbf{0} & \mathbf{C}_{TMDI-2} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{Y}}_1(t) \\ \dot{\mathbf{Y}}_2(t) \\ \dot{\mathbf{Y}}_{TMDI-1}(t) \\ \dot{\mathbf{Y}}_{TMDI-2}(t) \end{Bmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{0} & \gamma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 & \mathbf{0} & \gamma_2 \\ \gamma_1^T & \mathbf{0} & \mathbf{K}_{TMDI-1} & \mathbf{0} \\ \mathbf{0} & \gamma_2^T & \mathbf{0} & \mathbf{K}_{TMDI-2} \end{bmatrix} \begin{Bmatrix} \mathbf{Y}_1(t) \\ \mathbf{Y}_2(t) \\ \mathbf{Y}_{TMDI-1}(t) \\ \mathbf{Y}_{TMDI-2}(t) \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_1(t) \\ \mathbf{F}_2(t) \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix} \quad (7)$$

TMDI system can be regarded as an extension of the traditional TMD system, a special case of TMDI. Furthermore, when all parameters

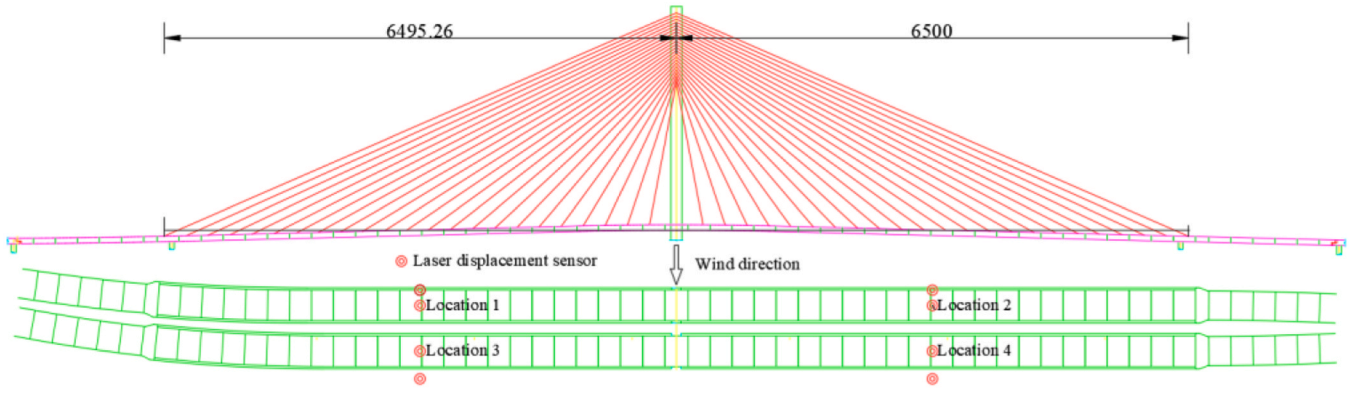


Fig. 4. Schematic layout of a (6495 + 6500) mm cable-stayed bridge with 1:50 scale (unit: mm).

related to the TMD are equal to zero in the second scheme, the equation degrades to the equation of motion of the original structure.

Thus, Eq. (2) can be written in the matrix form as follows

$$\mathbf{M}\ddot{\mathbf{Y}}(t) + \mathbf{C}\dot{\mathbf{Y}}(t) + \mathbf{K}\mathbf{Y}(t) = \mathbf{F}(t) \quad (9)$$

Owing to the influence of the equipped MTMDI device, the damping matrix are no longer orthogonal to the mode shapes. Hence, Eq. (9) can be transformed into a state-space form, which can be solved using a complex modal analysis method. Furthermore, this can be rewritten as a first-order differential equation (Eq. 10) with respect to the state vector

$$\dot{\mathbf{f}}(t) = \mathbf{A}\mathbf{z}(t) + \mathbf{B}\mathbf{z}(t) \quad (10)$$

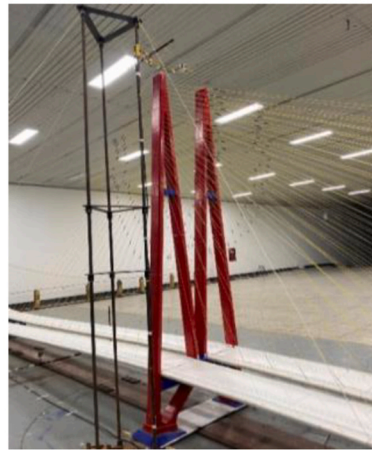
where $\mathbf{z}(t) = [\mathbf{Y}(t), \dot{\mathbf{Y}}(t)]^T$ denotes the state vector. The matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{f}(t)$ can be obtained as

$$\mathbf{A} = \begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix}, \quad \dot{\mathbf{f}}(t) = \begin{bmatrix} \mathbf{P}(t) \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{P}(t) = [\mathbf{F}_1(t) \quad \mathbf{F}_2(t)]^T \quad (11)$$

The aforementioned state-space form indicates that the transfer functions of the bridge system can be used to assess the effects of the

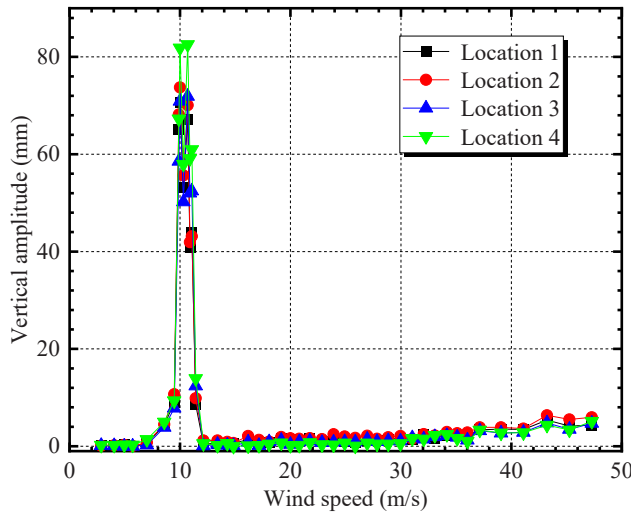


(a) the overall layout of the model

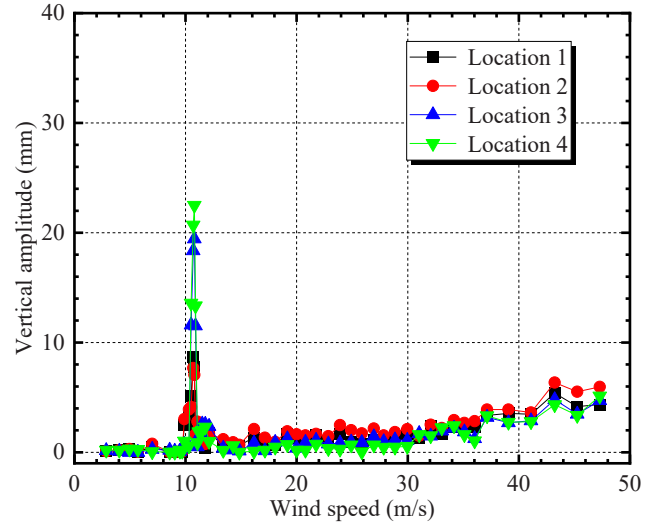


(b) detailed layout of the bridge tower and deck

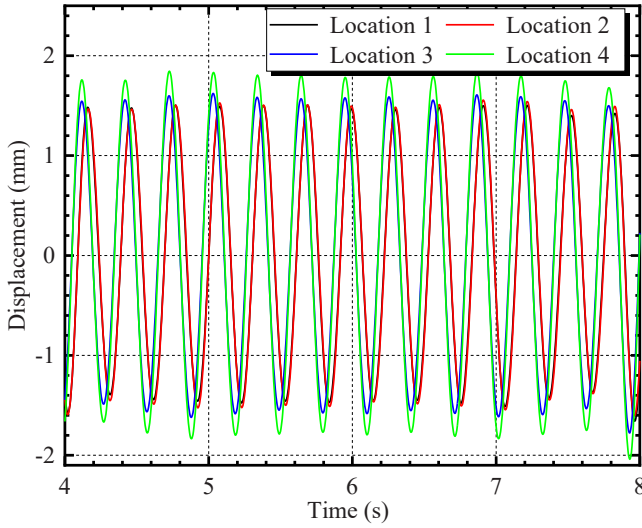
Fig. 5. Configuration of full bridge aeroelastic model test at the XNJD-3 wind tunnel.



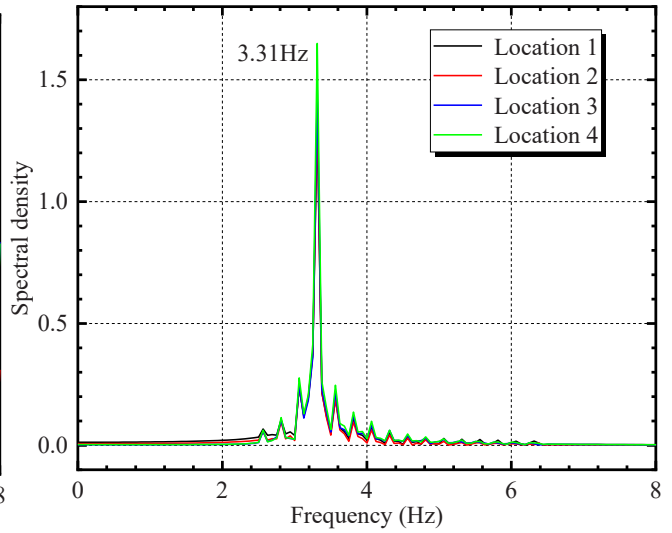
(a) vertical responses at 0° attack angle



(b) vertical responses at +3° attack angle



(c) time histories of vertical VIV at 0° attack angle



(d) spectrum of vertical VIV at 0° attack angle

Fig. 6. VIV characteristics of the original section.

attached TMDI on VIV suppression. The transfer functions of the displacement and acceleration responses at the p^{th} degree of freedom (DOF) generated by the forces at the q^{th} DOF can be obtained as follows

$$H_D^{pq}(j\omega) = \sum_{k=1}^N \left(\frac{\varphi_{pk}\varphi_{qk}}{a_k(j\omega - s_k)} + \frac{\varphi_{pk}^*\varphi_{qk}^*}{a_k^*(j\omega - s_k^*)} \right) \quad (12)$$

$$H_A^{pq}(j\omega) = \sum_{k=1}^N \left(\frac{-\omega^2 \varphi_{pk}\varphi_{qk}}{a_k(j\omega - s_k)} + \frac{-\omega^2 \varphi_{pk}^*\varphi_{qk}^*}{a_k^*(j\omega - s_k^*)} \right)$$

where φ_{pk} and a_k are the values of the p^{th} and q^{th} DOF, respectively, in the k^{th} complex mode shape of the complex modal analysis, respectively. In Eq. (12), a_k represents complex eigenvalue. The superscript * denotes the corresponding conjugate values. a_k and a_k^* are parameters that can be calculated as follows

$$\Phi^T \mathbf{A} \Phi = \text{diag} \{ a_1 \quad \cdots \quad a_k \quad a_1^* \quad \cdots \quad a_k^* \} \quad (13)$$

where Φ is a complex modal shape matrix that includes all eigenvectors ϕ_i . On the basis of the complex mode superposition method, the state-space vector response $\mathbf{z}(t)$ (under the assumption of a zero initial value) can be expressed as

$$\mathbf{z}(t) = \sum_{i=1}^{2n+2} \phi_i \mathbf{q}_i(t) \quad (14)$$

$$\mathbf{z}(t) = \sum_{i=1}^{n+1} \left(\frac{\phi_i}{a_i} \int_0^t F_i(\tau) e^{s_i(t-\tau)} d\tau + \frac{\phi_i^*}{a_i^*} \int_0^t F_i^*(\tau) e^{s_i^*(t-\tau)} d\tau \right)$$

where ϕ_i denotes the i^{th} mode shape in the real modal analysis, respectively. The i^{th} generalized force and its conjugate are expressed as

$$F_i(t) = \phi_i^T \mathbf{f}(t) \quad (15)$$

$$F_i^*(t) = \phi_i^{*T} \mathbf{f}(t)$$

Finally, the extreme responses are defined as

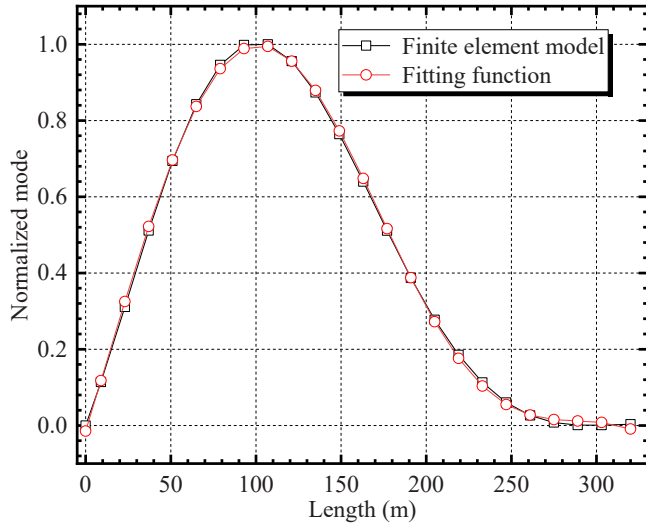


Fig. 7. Comparison of modal shape between fitting functions and FEM.

$$\hat{Z} = \bar{Z}(t) \pm g\sigma_z \quad (16)$$

where $\bar{Z}(t)$, σ_z , and g denote the average values, root mean square (RMS) values, and peak factor of the responses, respectively. For the VIVs with the sinusoidal property, g can be taken as $\sqrt{2}$. Generally, the average value of the VIV response is zero; thus, the RMS value is exclusively calculated.

3. Case studies

3.1. Wind tunnel tests of parallel bridges

3.1.1. Introduction of engineering background and wind tunnel

The deck section of the two parallel cable-stayed bridges with a single tower consists of bowl-shaped steel box girders. The main girders of the two bridges are 22.5 m wide with a gap distance of 6.5 m between their decks. Each bridge deck is equipped with sidewalk rails and traffic

barriers. The layout of the deck section is shown in Fig. 2.

The aerostatic tests were conducted in the XNJD-1 wind tunnel at Southwest Jiaotong University. The dimensions of experimental section are 2.4 m (width) $\times$ 2.0 m (height), and the wind speed range is 1.0–45.0 m/s with turbulence intensity being less than 0.5%. Aeroelastic model tests were conducted in the XNJD-3 wind tunnel at Southwest Jiaotong University. The cross-sectional dimensions of the tunnel are 22.5 m (width) $\times$ 4.5 m (height) $\times$ 36.0 m (length). The wind speeds range from 1.0 to 16.5 m/s. The turbulence characteristics, uniformity of the incoming flow speed, and other parameters satisfy the requirements of the aeroelastic model wind tunnel tests.

3.1.2. Aerostatic wind tunnel tests

The three components of the wind load per unit length acting on the bridge girder are the lift F_V , drag F_H , and pitch moment M_T , which can be written in the body axes as follows:

$$F_V = 0.5\rho U^2 C_v(\alpha) B \quad (17)$$

$$F_H = 0.5\rho U^2 C_H(\alpha) D \quad (18)$$

$$M_T = 0.5\rho U^2 C_M(\alpha) B^2 \quad (19)$$

where D is the vertical projected area of the girder. $C_v(\alpha)$, $C_H(\alpha)$, and $C_M(\alpha)$ are the lift, drag, and pitch moment coefficients in the body axes, respectively. α is the effective wind attack angle; and the other parameters are the same as those in Eq. (3). Correspondingly, the lift and drag coefficients on the wind axis are expressed as

$$C_L = C_v \cos(\alpha) - \frac{C_H D}{B \sin(\alpha)} \quad (20)$$

$$C_D = C_H \cos(\alpha) + \frac{C_v B}{D \sin(\alpha)} \quad (21)$$

where C_L and C_D are the lift and drag coefficients, respectively. Considering the blockage ratio of the wind tunnel, a 1:60 geometrical scale section model was produced and installed steadily on both sides, as shown in Fig. 3(a). The height, length and width of the model are 0.06 m, 2.095 m, and 0.375 m, respectively. With a force balance system installed, the wind attack angles can be adjusted accurately from

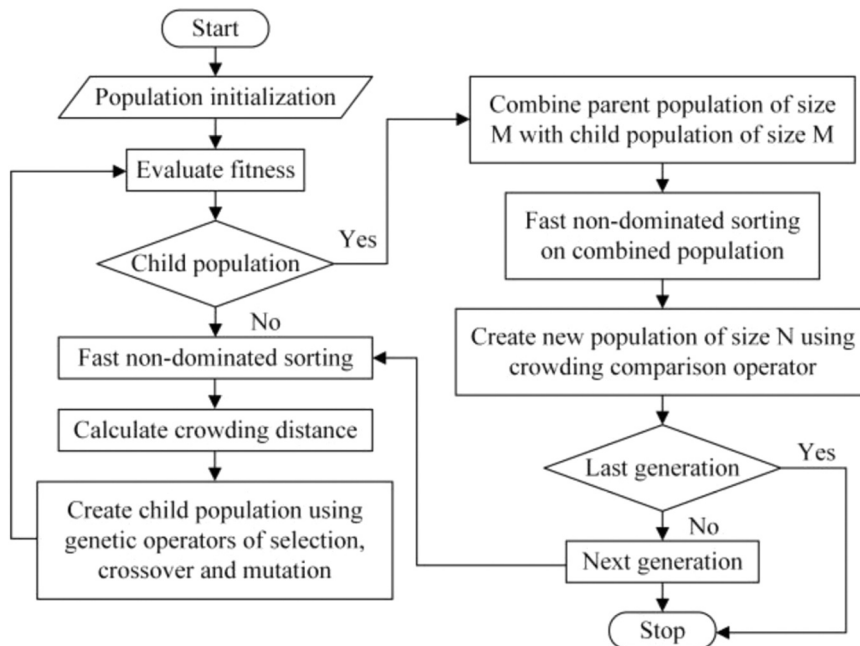


Fig. 8. The flow chart of NSGA-II.

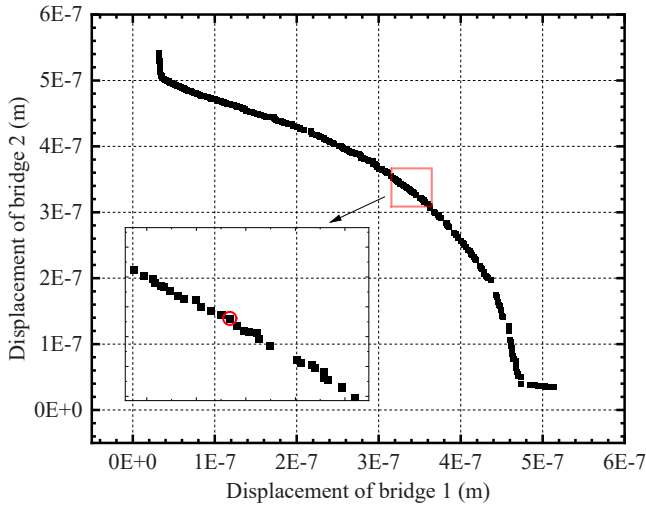


Fig. 9. Pareto front corresponding to the extreme displacement of Bridge 1 and Bridge 2.

Table 1

Design parameters of TMD and TMDI used for the comparative study.

Schemes	Parameters	Mass ratio (%)	Frequency ratio (%)	Damping ratio (%)	Inertance ratio (%)
2nd	TMD-1,2	0.3	99.7	3.35	-
3rd	TMDI-1	0.0965	100.00095	0.00044	0.0069067
3rd	TMDI-2	0.0972	100.00095	0.00064	0.0071254
4th	TMDI-1	0.0905	100.00094	0.00051	0.0000550
4th	TMDI-2	0.0904	100.00111	0.00063	0.0001260

Table 2

Masses of TMDs and TMDIs, and inertance of TMDIs.

Schemes	Parameters	Mass (kg)	Inertance (kg)
2nd	TMD-1,2	5098.2	-
3rd	TMDI-1	1639.9	117.4
3rd	TMDI-2	1651.8	121.1
4th	TMDI-1	1538.0	0.9
4th	TMDI-2	1536.3	2.1

– 6° to + 6° at intervals of 1°. A section model was tested at wind speeds of 8 m/s, 10 m/s, and 12 m/s in a smooth flow at 13 wind attack angles. The final static coefficients are obtained by averaging the results obtained at the above three wind speeds, under the condition that the difference does not exceed 2%. The test results are presented in Fig. 3(b).

3.1.3. Full bridge aeroelastic model wind tunnel tests

The schematic layout of the parallel bridges is shown in Fig. 4. The left and right spans of the parallel cable-stayed bridges, with approximate lengths of 325 m, are not symmetrical. The bridge tower is 167 m high. Displacement sensors are arranged at four locations below the deck to monitor the responses at different wind attack angles under a uniform flow. The layout of the 1:50 geometrical scale bridge model in the wind tunnel is shown in Fig. 5.

The damping ratios of the first order vertical mode and torsional mode of the aeroelastic model are adjusted to 0.47% and 0.53%, respectively, on the basis of free vibration decay tests. The frequency differences between the test values and theoretical design values (from the finite element model, FEM) remain less than 5%, which ensures the accuracy of the tests. At 0° wind attack angle, there are evident vertical VIVs at the four locations on the deck, as shown in Fig. 6(a). The wind

speed within lock-in region of the four locations is 8.5–12 m/s, and the amplitude at Location 4 is the largest, being 82.5 mm at a wind speed of 10.7 m/s. At + 3° wind attack angle, vertical VIVs are observed at four locations, as shown in Fig. 6(b). The wind speed within lock-in region of the four locations is 10.5–11 m/s, and the amplitude at Location 4 is the largest (22.5 mm) at a wind speed of 10.8 m/s. No vertical VIVs are observed at – 3° wind attack angle, and torsional VIVs are not observed at any of the three wind attack angles over the entire wind speed range. The results of these observations are not presented herein.

Fig. 6(c) depicts the time histories of the vertical displacement of the deck measured by the sensors in the wind tunnel at four locations, at a wind speed of 10.7 m/s. The displacement tends to be stable, and the displacement at Location 4 is significantly larger than that at the other locations. At 0° wind attack angle, the phase values at Locations 1 and 2 are significantly smaller than those at Locations 3 and 4. Similarly, Fig. 6(d) shows the spectral density values of the displacement time histories. The vibration frequency is 3.31 Hz, corresponding to the vertical frequency of the FEM, which is 0.4681 Hz. Thus, it can be concluded that first-order vertical VIVs of the deck exist at 0° wind attack angle. The time histories and spectral characteristics at + 3° attack angle are similar to those at 0° wind attack angle.

Based on the modal analysis results of the FEM, the mode shape of the vertical bending vibration of this order is shown in Fig. 7. The sine function using fitted mode shape is approximated as follows:

$$\varphi(x) = 0.747 \sin(0.0103x + 0.15) + 0.3257 \sin(0.02313x - 0.3991) \quad (22)$$

where x denotes the coordinates of the bridge in the span direction. As indicated by Fig. 7, for a single-tower cable-stayed bridge, the maximum mode of vibration does not occur at the midpoint of the span (162.5 m) but at a position approximately 100 m from one end, specifically at the 1/3 position of the span.

3.2. Optimal MTMDI parameters

3.2.1. Process of optimization

When the two individual bridge in the parallel bridges are modeled as a linear single-degree of freedom (SDOF) system respectively. In the fourth scheme, the eight parameters of the two TMDIs are the mass ratio $\mu_i = m_{\text{TMDI-}i}/M_i$, inertance ratio $\beta = b_i/M_i$ (where $i = 1, 2$ represent Bridge 1 and 2, b_i is the inertance of TMDI- i , and M_i is the total mass of Bridge i), frequency ratio $v_{\text{TMDI-}i} = \omega_{\text{TMDI-}i}/\omega_i$ (where $\omega_{\text{TMDI-}i}$ and ω_i are the natural frequencies of TMDI- i and the VIV frequency of Bridge i), and damping ratio $\zeta_{\text{TMDI-}i} = c_{\text{TMDI-}i}/(2\sqrt{\mu_{\text{eff}i}M_iK_{\text{TMDI-}i}})$ (where $\mu_{\text{eff}i} = \mu_i + \beta_i$, which is the effective mass ratio). Optimization to obtain the above-mentioned parameters was performed for the fourth scheme. The objective function entails the minimization of the extreme displacement responses of Bridges 1 and 2, subject to the constraints of $\mu_{\text{TMDI-}i}$, $v_{\text{TMDI-}i}$, $\beta_{\text{TMDI-}i}$, and $\zeta_{\text{TMDI-}i}$ in the fourth scheme. Similarly, because the TMDIs are separately installed in each bridge in the third scheme, the four parameters of the first TMDI-1— $\mu_{\text{TMDI-}1}$, $v_{\text{TMDI-}1}$, $\beta_{\text{TMDI-}1}$, and $\zeta_{\text{TMDI-}1}$ —must be exclusively optimized. The single-objective optimization problems (SOPs) in the third scheme (TMDI-1) are expressed in Eq. (23). The multi-objective optimization problems (MOPs) in the fourth scheme (TMDI-1,2) are expressed in Eq. (24).

$$\begin{cases} \text{minimize} & f_1(\mu_{\text{TMDI-}1}, \beta_{\text{TMDI-}1}, v_{\text{TMDI-}1}, \zeta_{\text{TMDI-}1}) = |\hat{D}_{\text{dist}}| \\ \text{s.t.} & \mu_{\text{TMDI-}1} \in [0.03\%, 0.1\%], \beta_{\text{TMDI-}1} \in [0, 1], \\ & v_{\text{TMDI-}1} \in [0.9, 1.1], \zeta_{\text{TMDI-}1} \in [0, 20\%]. \end{cases} \quad (23)$$

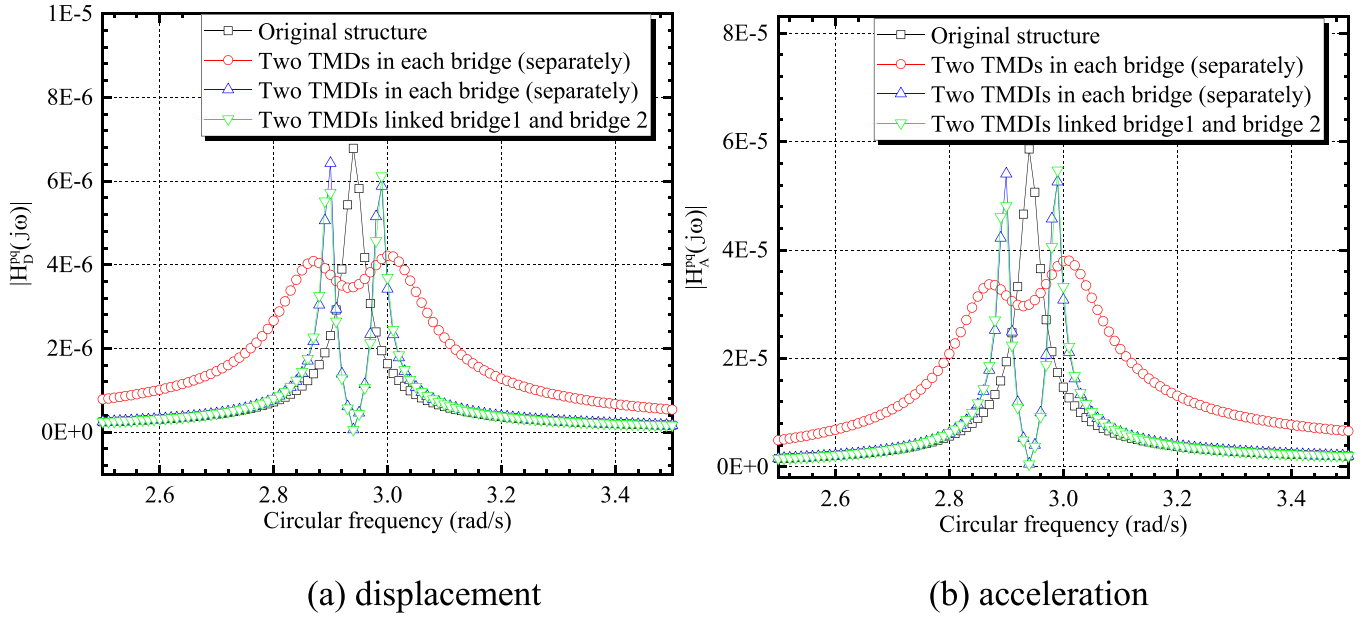


Fig. 10. Transfer functions of bridge 1 in different schemes.

$$\begin{aligned}
 &\text{minimize } f_1(\mu_{TMDI-1}, \beta_{TMDI-1}, v_{TMDI-1}, \zeta_{TMDI-1}) = |\hat{D}_{dis1}| \\
 &\left\{ \begin{aligned} &f_2(\mu_{TMDI-2}, \beta_{TMDI-2}, v_{TMDI-2}, \zeta_{TMDI-2}) = |\hat{D}_{dis2}| \\ &s.t. \quad \mu_{TMDI-1,2} \in [0.03\%, 0.1\%], \beta_{TMDI-1,2} \in [0, 1], \\ &\quad v_{TMDI-1,2} \in [0.9, 1.1], \zeta_{TMDI-1,2} \in [0, 20\%]. \end{aligned} \right. \quad (24)
 \end{aligned}$$

where $\hat{D}_{dis1}$ and $\hat{D}_{dis2}$ represent the extreme displacement responses of Bridges 1 and 2, respectively.

The load in the dynamic calculation is as shown in Eq. (3), and the parameters in the wind tunnel tests are converted into those of the actual bridge. The damping ratio of the bridge structure is taken as the common value, that is, 0.5% of the steel box girder according to the design code [51]. Considering that the natural vibration frequency of the structure is close to the vortex shedding frequency, both are taken as 0.4681 Hz. The phase differences between loads and responses are not considered for Bridge 1 and Bridge 2. In this study, the responses of the bridge structure are calculated using generalized coordinates. When the structure is in motion state, its displacement responses can be modally expanded and expressed as $y(t) = \sum_{i=1}^{\infty} \varphi_i(x) q_i(t)$. The results are integrated along the length L to obtain the equation at modal coordinate. In addition, the generalized stiffness and generalized damping matrix are employed to calculate these responses. $M_1^s = \bar{m} \int_0^L \varphi_i(x)^2 dx$ represents the modal mass for the i th mode; $\bar{m}$ denotes the equivalent mass of the deck per unit length, and L is the total length of Bridge 1. $\varphi_i(x)$ is the modal shape of the deck for the i th mode. The equivalent mass of the deck per unit length is 15848.6 kg/m. As a simplified harmonic force load model is adopted, the influence of aerodynamic stiffness and that of aerodynamic damping are ignored. However, the overall design of TMDI systems remains unaffected.

3.2.2. Results of optimization

The non-dominated sorting genetic algorithm (NSGA-II [52,53]) is used to solve MOPs, and the flow chart is shown in Fig. 8. As shown in the previous section, the objective function of MOPs is the displacement responses of Bridge 1 and Bridge 2. The optimization needs to be carried out under specified constraint conditions (i.e., Eq. (24)). After adjusting the parameters, the evolution generation is 400, and the number of populations in each generation is 300. A Pareto front can be obtained to minimize the extreme displacement of Bridges 1 and 2 in the fourth

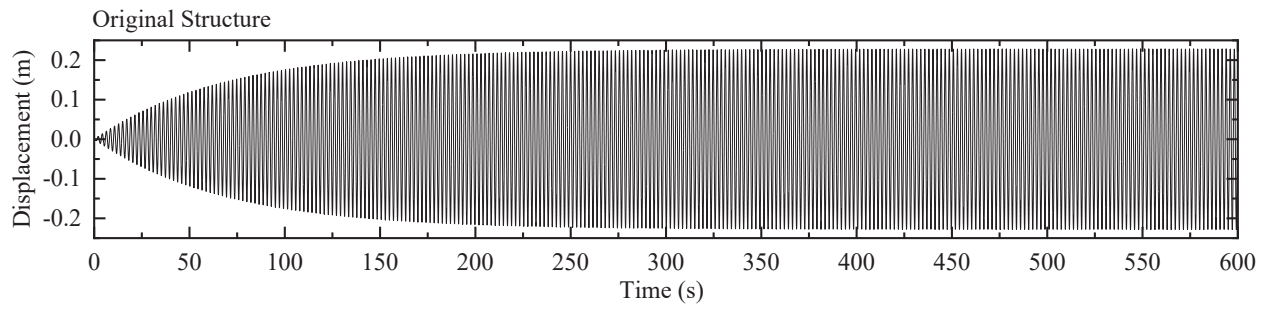
scheme, as shown in Fig. 9. In Fig. 9, all points are optimal, and the point inside the red circle is a representative. At this point, the response values obtained by the two targets are equal.

All eight optimized parameters in the fourth scheme of the TMDIs are obtained from the Pareto front and are listed in Table 1. The four optimized TMDI parameters in the third scheme are included in Table 1. The parameters used for the TMD are listed in Table 1 so as to compare the performances of the TMD and TMDI. The mass ratio of the TMD for the bridge structure is less than 1% due to space limitations. For example, in the Hong Kong-Zhuhai-Macao Bridge, the mass of the installed TMD is 32 t, accounting for 0.36% of the mass of the structure, based on the finite element analysis results [27]. Therefore, a mass ratio of 0.3% is used in this study. A preliminary estimate of the frequency ratio and damping ratio of the TMD can be calculated using the formula as follows suggested in the design code [51], where μ_{TMD} is the mass ratio of TMD device.

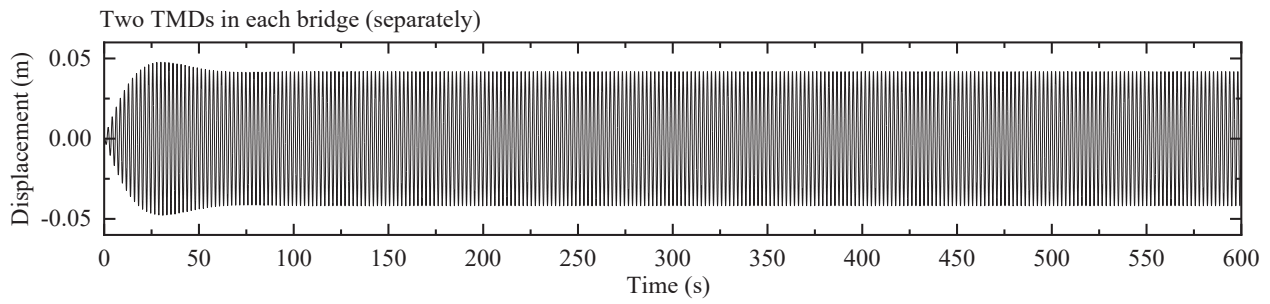
$$v_{TMDI} = \frac{1}{1 + \mu_{TMD}} \quad (25)$$

$$\zeta_{TMDI} = \sqrt{\frac{3\mu_{TMD}}{8(1 + \mu_{TMD})}} \quad (26)$$

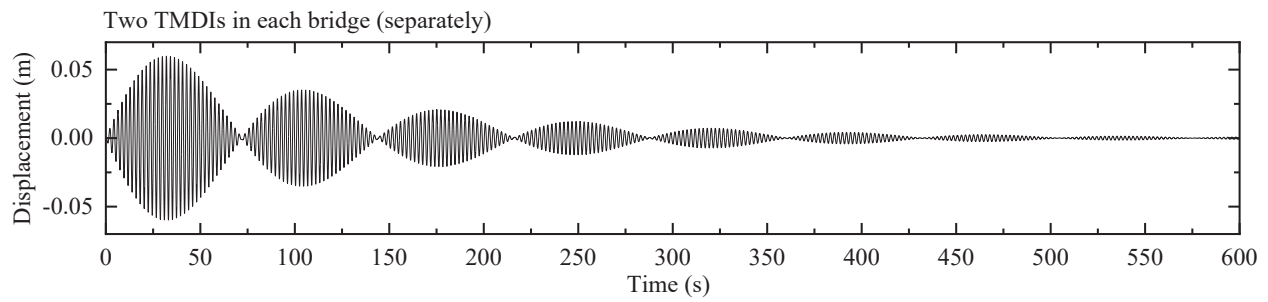
The optimization results demonstrate that there is no significant difference in the structural response values between the third and fourth schemes, wherein the TMDI is installed. Furthermore, the response values are significantly reduced. As shown in Table 1, the mass ratio, frequency ratio, and damping ratio of the third and fourth schemes do not exhibit significant differences. However, the inertance ratio of the third scheme is larger than that of the fourth scheme. Specifically, as listed in Table 2, in the fourth scheme, the masses of TMDI-1 and TMDI-2 are 1538.0 kg and 1536.3 kg, respectively, while the masses are 1639.9 kg and 1651.8 kg, respectively, in the third scheme. The mass of the TMD is 5098.2 kg in the second scheme. In the fourth scheme, the inertances of TMDI-1 and TMDI-2 are 0.9 kg and 2.1 kg, respectively. In comparison, the inertances of TMDI-1 and TMDI-2 are 117.4 kg and 121.1 kg, respectively, in the third scheme. The inertance represents the equivalent mass produced by the inerter, rather than the actual mass. Therefore, the fourth scheme is superior in terms of meeting the inertance requirements and reducing the costs associated with the TMDI.



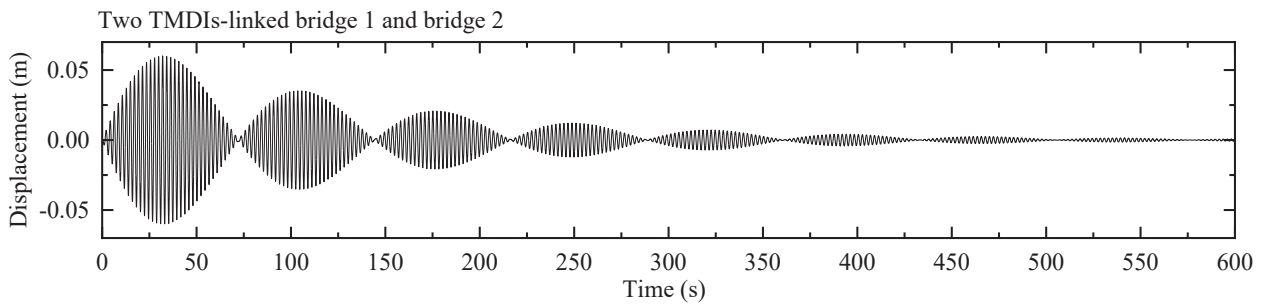
(a) original structure



(b) two TMDs in each bridge (separately)



(c) two TMDIs in each bridge (separately)



(d) two TMDIs-linked bridge 1 and bridge 2

Fig. 11. Time histories of displacement of the four schemes.

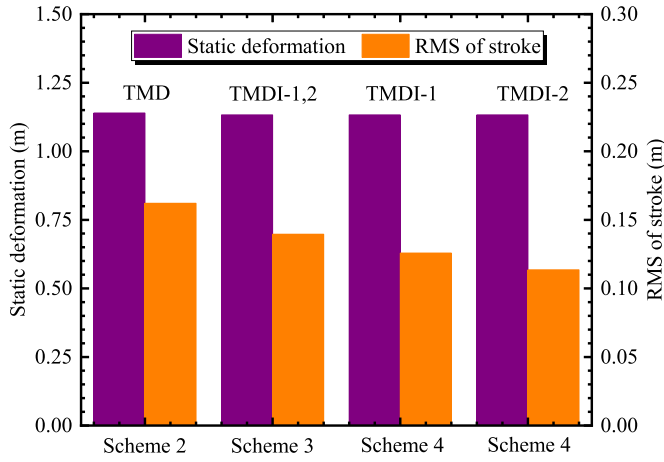


Fig. 12. Static stretching and stroke of the TMD and TMDI.

3.3. Effectiveness of the MTMDI in VIV mitigation

3.3.1. Comparison of transfer functions

Once the parameters are obtained, the displacement and acceleration transfer functions are determined using Eq. (12), as depicted in Fig. 10, which help estimate the effects of the TMDI in mitigating wind-induced responses. Figs. 10(a) and 10(b) illustrate the displacement and acceleration transfer functions of Bridge 1 in terms of four schemes. Evidently, the peak values of the displacement and acceleration transfer functions around the first order natural frequency decrease significantly when using the TMDI (blue and green lines) compared with the TMD (red line) and the original structure (black line). As shown in Figs. 10(a) and 10(b), one peak in the original structure is replaced by two peaks distributed beside the natural frequency after the installation of the TMDI or TMD. The peak values of the displacement and acceleration transfer functions at the natural frequency of the bridge are reduced from 6.8×10^{-6} and 6.0×10^{-5} , respectively, to 0. VIVs are less likely to occur when the vibration frequency of the structure does not match the vortex shedding frequency.

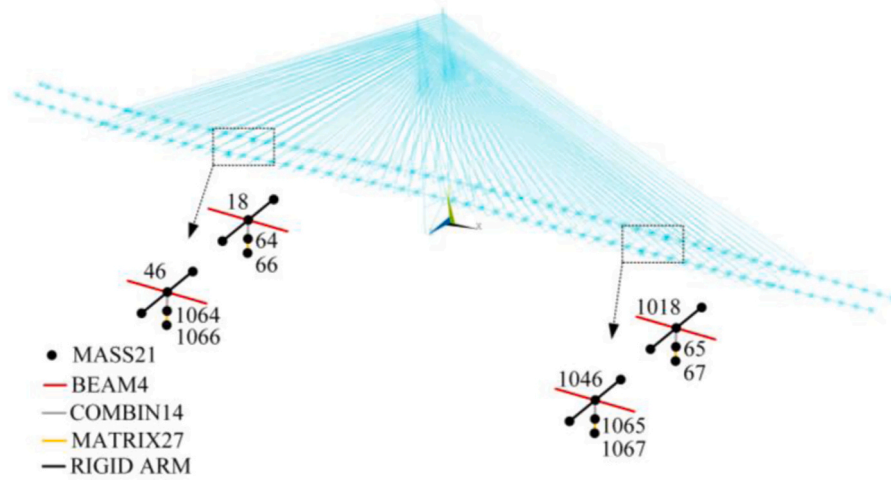


Fig. 13. Implementation of TMDs and TMDIs in the FEM of the bridge.

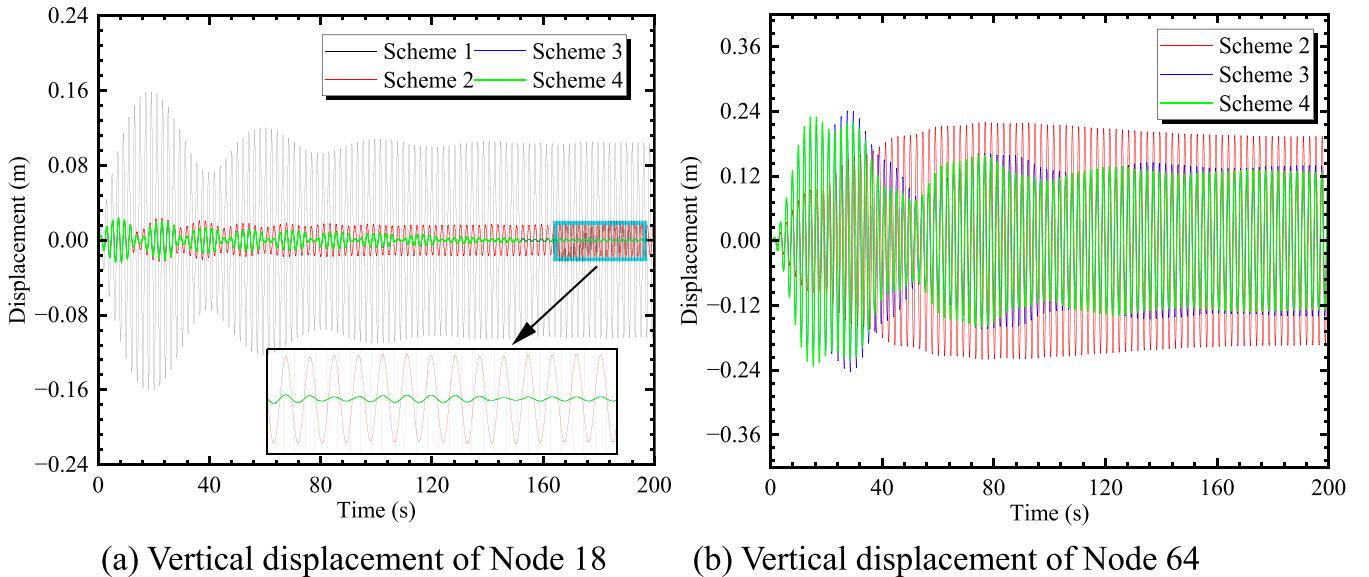


Fig. 14. Time histories of displacement during vibration control.

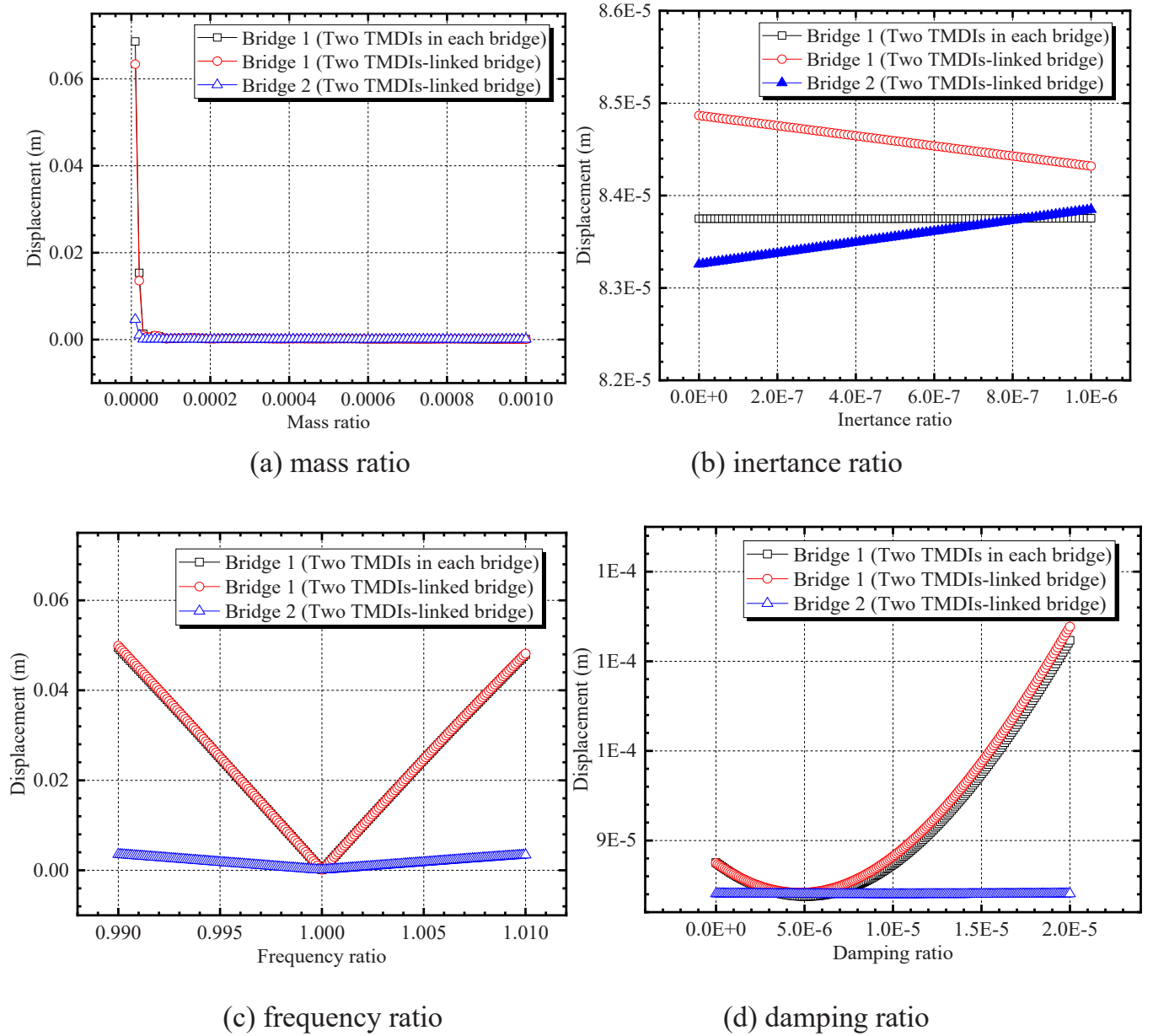


Fig. 15. Influence of one-parameter change on bridge response.

3.3.2. Comparison of displacement time histories

The displacement time histories of Bridges 1 and 2 are obtained by substituting the optimized MTMDI and TMD parameters into the dynamic motion equation. Fig. 11 shows the displacement time histories of Bridge 1 for the four schemes. In the first scheme, the displacement initially demonstrates an increasing tendency with time and reaches a stable maximum response value as the load continues. Within the full range of time histories, the displacement response values do not exceed 0.25 m. In the second scheme, the displacement also increases with time but begins to decrease after reaching a maximum value of approximately 0.05 m. The stable tendency is sustained in the following time, indicating the insignificant mitigation effects of the TMD. The displacement values in the third and fourth schemes are highly similar, suggesting a continuous increase at the beginning, with the maximum value not exceeding 0.07 m. The similarity of displacement values arises from the fact that the minimized displacement of two schemes serves as the object in multi-objective optimization. Although the optimized parameters of the two schemes are different, both have achieved the effect of reducing

the VIV displacement responses. This is also reflected in the displacement transfer functions of the two schemes, which are also similar. However, after a period of continuous oscillations, the response values decrease until the vibration eventually disappears, achieving the effect of restraining the VIVs of Bridge 1. The gradual decrease of the response values is consistent with the findings of a previous study [40].

Notably, the lift coefficients of the windward and leeward girders are not the same, resulting in different MTMDI parameters after optimization, particularly in the fourth scheme with complex coupled motion. When the direction of the incoming flow changes, that is, when the windward and leeward girders exchange positions, the same parameters are used to compare the control effects of the displacement responses, and the displacement responses can still be attenuated. This indicates that the application of MTMDI systems to the VIV control of parallel bridges meets the expectations.

3.3.3. Feasibility analysis and performances evaluation

The height limitation of the main girder may impede the practical

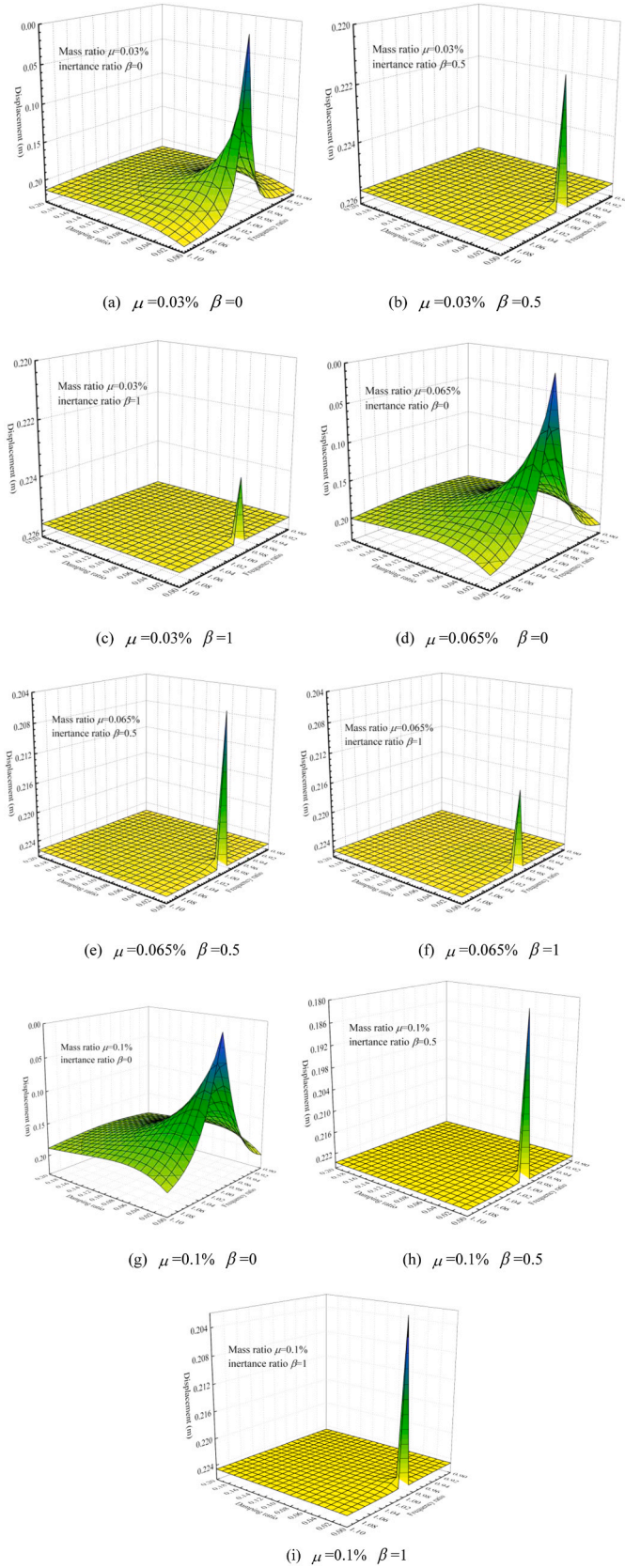


Fig. 16. Influence of mass ratio and inertance ratio on the intensity contour of the main girder displacement.

application of TMDIs. To verify the feasibility of installing TMDI devices on real bridges, Fig. 12 presents the static stretching and RMS values of the dynamic stroke for different devices. Static stretching is calculated by $g/\omega_{TMD\ or\ TMDI-i}^2$, where g is the gravity acceleration. The RMS values of the dynamic stroke are obtained using numerical calculations. Fig. 12 indicates that the third and fourth schemes have the same static stretching, reaching 1.1329 m, which is smaller than that of the second scheme (1.1397 m). Regarding the RMS values of the dynamic stroke, the third (0.1395 m) and fourth schemes (TMDI-1: 0.1258 m, TMDI-2: 0.1136 m) exhibit values that are 13.9%, 22.4%, and 29.9% smaller than that of the second scheme (0.1621 m), respectively. It can be concluded that the dynamic stroke of TMDI is smaller to that of TMD.

FEM analysis provides more accurate and comprehensive results, as it considers the effects of mode shapes, towers, cables, etc. The feasibility of the TMDI is further validated through time domain calculation. FEMs of the installation of TMDs, TMDIs, and TMDIs-linked, respectively, are established using ANSYS. Specifically, the second, third, and fourth schemes are modeled. The main girder and the towers are modeled by BEAM4. The cables are modeled by LINK8. The springs and dampers are modeled by COMBIN14 element, and the inerter is modelled by MATRIX27 element. Mass blocks are modeled using MASS21 element. Mass blocks are connected to the node with the largest mode shape value. In the third scheme, one end of the inerter is linked to the other bridge by a rigid arm, as shown in Fig. 13.

The transient analysis was performed with the same load as that in the previous simplified model, based on the parameters optimized by the genetic algorithm. The displacement time histories of Node 18 and Node 64 are depicted in Fig. 14(a) and (b), respectively. The calculated results are nearly identical to those of the simplified model, demonstrating a better control effect of the TMDIs and TMDIs-linked. The installation of the TMDIs makes the bridge no longer vibrate, and TMDIs vibrate continuously, thereby achieving the effect of VIV mitigation. The accuracy of the numerical calculation using the simplified model is further verified in terms of VIVs with a single degree of freedom.

3.4. Sensitivity analysis of the MTMDI systems

3.4.1. Single parameter analysis

To evaluate the effects of different MTMDI parameters on the VIV responses of the bridge, Fig. 15 elaborates on the variations in the maximum responses of Bridge 1 in the third scheme and both bridges in the fourth scheme when one parameter of TMDI-1 changes. In addition to the parameter that should be altered, other parameters remain optimized. Fig. 15(a) presents the variations in the maximum displacement with the mass ratio, where it is observed that the response values do not decrease significantly as the mass ratio increases to a certain value. This illustrates the insignificant suppression effects of the MTMDI on VIVs when the mass ratio remains small, whereas the effects improve with an increase in the mass ratio. Fig. 15(b) shows the variations in the displacement response values with the inertance ratio, which demonstrates that the structural responses remain relatively unchanged when the inertance ratio increases in the third scheme. In the fourth scheme, an increase in the inertance ratio of TMDI-1 leads to a decrease in the responses of Bridge 1 and an increase in the responses of Bridge 2. According to the comparison between the frequency ratio and response values of the bridge in Fig. 15(c), the responses of Bridge 1 are significantly influenced when the frequency ratio of TMDI-1 changes, both in the third and fourth schemes. By contrast, the responses of Bridge 2 are slightly affected by the variations in the frequency ratio of TMDI-1 in the fourth scheme. Therefore, when designing a MTMDI, attention should be paid to the frequency ratio. The relationship between the damping ratio and responses in Fig. 15(d) demonstrates a tendency in which the responses of Bridge 1 start small and then increase continually, whereas the responses of Bridge 2 are hardly affected when the damping ratio of TMDI-1 increases, both in the third and fourth schemes.

3.4.2. Two parameters analysis

For further investigation into the maximum responses of the parallel bridges when the two parameters of the MTMDI change simultaneously, the fourth scheme is considered as an example. Generally, in the implementation of a MTMDI, its mass and inertance remain fixed, making it reasonable to vary the frequency ratio and damping ratio. In this study, the mass ratios are set to 0.03%, 0.065%, and 0.1%, while the inertia coefficient ratios are set to 0, 0.5, and 1. Figs. 16(a), (d), and (g) show that the responses of the structure are significantly affected by the damping ratio and frequency ratio when the TMDI system degenerates into a TMD system, that is, when the inertance ratio is zero. With the appropriate damping ratio and frequency ratio of the TMD, the vibration responses significantly decrease. The higher the mass ratio of the TMD, the better the mitigation effects. In Fig. 16, in terms of the MTMDI system where the inertance ratio is not zero, variations in the damping ratio and frequency ratio exert little influence on the structural responses compared with the same TMDI mass ratio. In addition, when the frequency ratio is close to 1 and the damping ratio is low, the responses are smaller than those under other parameters. However, for the same inertance ratio, there is no apparent correlation between the mass ratio of the TMDI and the responses. For example, when the inertance ratio is 0.5 in Fig. 16(b), (e), and (h), smaller responses are observed with an increase in mass ratios. However, in Fig. 16(c), (f), and 16(i), when the inertance ratio is 1, the responses do not exhibit regularity with an increase in mass ratios. In conclusion, the MTMDI system is superior to the TMD system in terms of robustness because variations in the MTMDI parameters do not lead to excessive changes in the structural responses. However, the frequency ratio of the MTMDI significantly affects the structural responses. Therefore, in an actual design, the VIV frequency of the structure should be accurately determined according to which the TMDI parameters can be designed to suppress VIVs more efficiently.

4. Conclusions

In the present study, an analytical model equipped with the MTMDI was proposed by simplifying a parallel bridge under VIVs into a SDOF system. The main conclusions are as follows.

First, first-order vertical VIVs at 0° and $+3^\circ$ wind attack angles were observed through full bridge aeroelastic wind tunnel tests. At 0° attack angle, the lock-in range and amplitudes of the VIVs are relatively the same for the windward and leeward bridges. However, the vibration phase on the windward side lags behind that on the leeward side. At $+3^\circ$ attack angle, the lock-in range is the same; however, the amplitude is greater on the leeward side than on the windward side.

Second, the optimized MTMDI parameters demonstrate superior suppression effects of the linked MTMDI (the fourth scheme) compared to the separate installation of TMDs and TMDIs (the second and third schemes), as the MTMDI provides a cost-effective passive vibration control solution with a small mass ratio and inertance ratio.

Third, the TMDIs and TMDs systems tune the single frequency (one-peak characteristic) of the original structure to two frequencies (two-peak characteristic), resulting in a decrease in the values of the transfer function. The values of the displacement and acceleration transfer functions at the natural frequency of the bridge are reduced from 6.8×10^{-6} and 6.0×10^{-5} , respectively, to 0, thus suppressing VIVs.

Furthermore, the displacement time histories suggest that the suppression effects of the TMDs are limited. In contrast, the inerter-based passive vibration control system, that is the MTMDI system, demonstrates a more effective performance in suppressing the VIVs in terms of numerical calculations and evaluation of FEM.

Finally, it is concluded that the frequency of the MTMDI system is more sensitive than other parameters, which should be emphasized in engineering design. Meanwhile, the mass ratio, inertance ratio, and damping ratio exhibit less sensitivity to structural responses.

CRedit authorship contribution statement

Lei Wei: Conceptualization, Investigation, Methodology, Validation, Writing – original draft. **Wang Qi:** Data curation, Funding acquisition, Investigation, Validation, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Yang F, Zheng S, Yan Z. Vortex-induced vibration and control of split three-box girder bridges. *Struct Eng Int* 2021;1–10.
- [2] Zhou Q, Zheng S, Chen Z, Yang F, Liu J. Vortex-induced vibration reliability analysis in cable-stayed bridges: wind and aerodynamic uncertainties. *Struct Eng Int* 2021;31(1):51–9.
- [3] Gao DL, Deng Z, Yang WH, Chen WL. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: a vortex-dynamics approach. *J Fluids Struct* 2021;105:103348.
- [4] Li K, Qian GW, Ge YJ, Zhao L, Di J. Control effect and mechanism investigation on the horizontal flow-isolating plate for PI shaped bridge decks' VIV stability. *Wind Struct* 2019;28(2):99–110.
- [5] Zhu Q, Xu YL, Zhu LD, Li H. Vortex-induced vibration analysis of long-span bridges with twin-box decks under non-uniformly distributed turbulent winds. *J Wind Eng Ind Aerodyn* 2018;172:31–41.
- [6] Nagao F, Utsunomiya H, Yoshioka E, Ikeuchi A, Kobayashi H. Effects of handrails on separated shear flow and vortex-induced oscillation. *J Wind Eng Ind Aerodyn* 1997;71:819–27.
- [7] Buljac A, Kozmar H, Pospíšil S, Macháček M, Kuznetsov S. Effects of wind-barrier layout and wind turbulence on aerodynamic stability of cable-supported bridges. *J Bridge Eng* 2020;25(12):04020102.
- [8] Laima S, Li H, Chen W, Ou J. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J Fluids Struct* 2018;77:115–33.
- [9] Li M, Sun YG, Jing HM, Li MS. Vortex-induced vibration optimization of a wide streamline box girder by wind tunnel test. *KSCE J Civ Eng* 2018;22(12):5143–53.
- [10] Hu C, Zhao L, Ge Y. Mechanism of suppression of vortex-induced vibrations of a streamlined closed-box girder using additional small-scale components. *J Wind Eng Ind Aerodyn* 2019;189:314–31.
- [11] Zhou R, Ge Y, Liu Q, Yang Y, Zhang L. Experimental and numerical studies of wind-resistance performance of twin-box girder bridges with various grid plates. *Thin Walled Struct* 2021;166.
- [12] Ma C, Li Z, Meng F, Liao H, Wang J. Wind-induced vibrations and suppression measures of the Hong Kong-Zhuhai-Macao Bridge. *Wind Struct* 2021;32(3):179–91.
- [13] Liu JS, Wang F, Yang Y, Liu RH. Nonlinear flutter analysis for a long-span suspension bridge. *Shock Vib* 2021;2021:5572429.
- [14] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. *Struct Eng Int* 2013;23(3):248–68.
- [15] Liu SY, Zhao L, Fang GS, Hu CX, Ge YJ. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. *J Fluids Struct* 2021;106:103380.
- [16] Li H, Laima SJ, Zhang QQ, Li N, Liu ZQ. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2014;124:54–67.
- [17] Bai H, Ji N, Xu G, Li J. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: Sealed traffic barrier. *J Wind Eng Ind Aerodyn* 2020;206:104302.
- [18] Chen XH, Kareem A. Efficacy of tuned mass dampers for bridge flutter control. *J Struct Eng* 2003;129(10):1291–300.
- [19] Dai J, Xu ZD, Gai PP. Parameter determination of the tuned mass damper mitigating the vortex-induced vibration in bridges. *Eng Struct* 2020;221:111084.

- [20] Chang CC, Gu M, Tang KH. Tuned mass dampers for dual-mode buffeting control of bridges. *J Bridge Eng* 2003;8(4):237–40.
- [21] Sun Z, Zou ZL, Ying XY, Li XQ. Tuned mass dampers for wind-induced vibration control of Chongqi Bridge. *J Bridge Eng* 2020;25(1):05019013.
- [22] Wang WX, Wang XY, Hua XG, Song GB, Chen ZQ. Vibration control of vortex-induced vibrations of a bridge deck by a single-side pounding tuned mass damper. *Eng Struct* 2018;173:61–75.
- [23] Kim S, Park J, Kim HK. Damping identification and serviceability assessment of a cable-stayed bridge based on operational monitoring data. *J Bridge Eng* 2017;22(3):04016123.
- [24] Nagarajaiah S. Adaptive passive, semiactive, smart tuned mass dampers: identification and control using empirical mode decomposition, hilbert transform, and short-term fourier transform. *Struct Control Health Monit* 2009;16(7–8): 800–41.
- [25] Hoang N, Fujino Y, Warnitchai P. Optimal tuned mass damper for seismic applications and practical design formulas. *Eng Struct* 2008;30(3):707–15.
- [26] Li CX, Li QS. Evaluation of the lever-type multiple tuned mass dampers for mitigating harmonically forced vibration. *Int J Struct Stab Dyn* 2005;5(4):641–64.
- [27] Dai J, Xu ZD, Yin XJ, Gai PP, Luo Y. Parameters design of TMD mitigating vortex-induced vibration of the Hong Kong-Zhuhai-Macao bridge deep-water nonnavigable bridge. *J Bridge Eng* 2019;24(8):06019005.
- [28] De Domenico D, Ricciardi G. An enhanced base isolation system equipped with optimal tuned mass damper inerter (TMDI). *Earthq Eng Struct Dyn* 2018;47(5): 1169–92.
- [29] Pietrosanti D, De Angelis M, Basili M. Optimal design and performance evaluation of systems with Tuned Mass Damper Inerter (TMDI). *Earthq Eng Struct Dyn* 2017; 46(8):1367–88.
- [30] Marian L, Giaralis A. Optimal design of a novel tuned mass-damper-inerter (TMDI) passive vibration control configuration for stochastically support-excited structural systems. *Probab Eng Mech* 2014;38:156–64.
- [31] Petrini F, Giaralis A, Wang ZX. Optimal tuned mass-damper-inerter (TMDI) design in wind-excited tall buildings for occupants' comfort serviceability performance and energy harvesting. *Eng Struct* 2020;204:109904.
- [32] Giaralis A, Petrini F. Wind-induced vibration mitigation in tall buildings using the tuned mass-damper-inerter. *J Struct Eng* 2017;143(9):04017127.
- [33] Giaralis A, Taflanidis AA. Optimal tuned mass-damper-inerter (TMDI) design for seismically excited MDOF structures with model uncertainties based on reliability criteria. *Struct Control Health Monit* 2018;25(2):e2082.
- [34] De Domenico D, Ricciardi G. Optimal design and seismic performance of tuned mass damper inerter (TMDI) for structures with nonlinear base isolation systems. *Earthq Eng Struct Dyn* 2018;47(12):2539–60.
- [35] Zhu ZW, Lei W, Wang QH, Tiwari N, Hazra B. Study on wind-induced vibration control of linked high-rise buildings by using TMDI. *J Wind Eng Ind Aerodyn* 2020; 205:104306.
- [36] MA Ruisheng, Bi K, Hao H. Inerter-based structural vibration control: a state-of-the-art review. *Eng Struct* 2021;243:112655.
- [37] Qiao H, Huang P, De DD, Wang Q. Structural control of high-rise buildings subjected to multi-hazard excitations using inerter-based vibration absorbers. *Eng Struct* 2022;266:114666.
- [38] Dai J, Xu ZD, Gai PP. Tuned mass-damper-inerter control of wind-induced vibration of flexible structures based on inerter location. *Eng Struct* 2019;199: 109585.
- [39] Wang ZX, Giaralis A. Enhanced motion control performance of the tuned mass damper inerter through primary structure shaping. *Struct Control Health Monit* 2021;28(8):e2756.
- [40] Xu K, Bi KM, Han Q, Li XP, Du XL. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: analytical study. *Eng Struct* 2019; 182:101–11.
- [41] Xu K, Dai Q, Bi KM, Fang GS, Ge YJ. Closed-form design formulas of TMDI for suppressing vortex-induced vibration of bridge structures. *Struct Control Health Monit* 2022:e3016.
- [42] Wang QH, Tiwari ND, Hazra B, Lei W, Zhu ZW. MTMDI for mitigating wind induced responses of linked high-rise buildings. *J Struct Eng* 2021;147(4): 06021001.
- [43] De Domenico D, Qiao HS, Wang QH, Zhu ZW, Marano G. Optimal design and seismic performance of multi-tuned mass damper inerter (MTMDI) applied to adjacent high-rise buildings. *Struct Des Tall Spec Build* 2020;29(14):e1781.
- [44] Dai J, Xu ZD, Gai PP, Hu ZW. Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges. *Mech Syst Signal Process* 2021;148:107180.
- [45] Xu K, Dai Q, Bi K, Fang G, Zhao L. Multi-mode vortex-induced vibration control of long-span bridges by using distributed tuned mass damper inerters (DTMDIs). *J Wind Eng Ind Aerodyn* 2022;224:104970.
- [46] Kim SJ, Kim HK, Calmer R, Park J, Kim GS, Lee DK. Operational field monitoring of interactive vortex-induced vibrations between two parallel cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2013;123:143–54.
- [47] Seo JW, Kim HK, Park J, Kim KT, Kim GN. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. *J Wind Eng Ind Aerodyn* 2013;116: 7–20.
- [48] Li C, Liang M, Wang YX, Dong YT. Vibration suppression using two-terminal flywheel. Part I: Modeling and characterization. *J Vib Control* 2012;18(8): 1096–105.
- [49] Smith MC. Synthesis of mechanical networks: the inerter. *IEEE Trans Autom Control* 2002;47(10):1648–62.
- [50] Nikoo HM, Bi KM, Hao H. Passive vibration control of cylindrical offshore components using pipe-in pipe (PIP) concept: An analytical study. *Ocean Eng* 2017; 142:39–50.
- [51] JTG/T 3360-01-2018. Wind-resistant design specification for highway bridges. Beijing: China: Communications Press; 2018.
- [52] Li H, Zhang QF. Multiobjective optimization problems with complicated pareto sets, MOEA/D and NSGA-II. *IEEE Trans Evol Comput* 2009;13(2):284–302.
- [53] Deb K, Pratap A, Agarwal S, Meyarivan T. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans Evol Comput* 2002;6(2):182–97.